AUTOM-03: Monaco Configuration-as-Code

> **Series:** AUTOM — Dynatrace Automation | **Notebook:** 3 of 9 | **Created:** January 2026 | **Last Updated:** 07/31/2026

Monaco (Monitoring as Code) is Dynatrace's official CLI tool for configuration management. It uses YAML files to define configurations and supports version control, CI/CD integration, and multi-environment deployments.

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Prerequisites

Before starting this notebook, ensure you have:

Requirement	Description
Monaco CLI	Installed via GitHub releases or brew
API Token	Token with `settings.read`, `settings.write`, `ReadConfig`, `WriteConfig`

Requirement	Description
Tenant URL	Your Dynatrace SaaS tenant URL

Learning Objectives

By the end of this notebook, you will:

- Understand Monaco's configuration-as-code model
- Know how to download existing configurations
- Be able to create and deploy configurations via YAML
- Handle multi-environment deployments

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1. Introduction

Why Monaco?

Benefit	Description
Version Control	Track all changes in Git
Repeatability	Deploy same config to multiple environments
Review Process	Use pull requests for config changes
Disaster Recovery	Quickly restore configurations
Documentation	YAML files serve as living documentation

Monaco vs Direct API

Aspect	Monaco	Settings API
Format	YAML files	JSON payloads
State management	Implicit (by name)	Manual tracking

Aspect	Monaco	Settings API
Multi-tenant	Built-in support	Custom scripting
Dry run	Yes (`--dry-run`)	No
Delete orphans	Yes (`delete` command)	Manual

> **Already a Terraform shop?** See **AUTOM-01 §5: When a Terraform Shop Should Add Monaco** for the five patterns where Monaco still earns its place — most commonly bulk download from an existing tenant, multi-tenant identical-config deployment, and app-team self-service without state management.

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2. Getting Started

Installation

macOS (Homebrew):

```
brew install dynatrace-oss/monaco/monaco
```

Linux/Windows:

```
# Download from GitHub releases
curl -L https://github.com/Dynatrace/dynatrace-configuration-as-
code/releases/latest/download/monaco-linux-amd64 -o monaco
chmod +x monaco
```

Verify installation:

```
monaco version
```

Environment Setup

Create environment variables for authentication:

```
export DT_TENANT_URL="https://{tenant-id}.live.dynatrace.com"
export DT_API_TOKEN="&lt;your-api-token&gt;"
```

Your First Download

Download all configurations from your tenant:

```
monaco download \
  --environment-url "$DT_TENANT_URL" \
  --api-token "$DT_API_TOKEN" \
  --output-folder ./downloaded-config
```

Download specific configuration types:

```
monaco download \
  --environment-url "$DT_TENANT_URL" \
  --api-token "$DT_API_TOKEN" \
  --output-folder ./downloaded-config \
  --api builtin:management-zones,builtin:tags.auto-tagging
```

> **Downloading classic schemas is the point here, not an oversight.** Both schemas above are marked **Blocked at upgrade** in AUTOM-02's Settings 2.0 catalog — they work today and stop answering once your tenant moves to the latest Dynatrace. That makes `monaco download` the fastest way to *inventory* what you still have before migrating it: every object, in version control, in one command. Keep the export as your migration worksheet; retarget the deployment. What it should not become is a template for new configuration.

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3. Project Structure

Recommended Layout

```
dynatrace-config/
├── manifest.yaml              # Project manifest
├── environments/
│   ├── development.yaml       # Dev environment config
│   ├── staging.yaml           # Staging config
│   └── production.yaml        # Production config
└── projects/
    └── my-project/
        ├── management-zones/
        │   └── config.yaml
        ├── auto-tagging/
        │   └── config.yaml
        └── alerting-profiles/
            └── config.yaml
```

Manifest File

The `manifest.yaml` defines your project:

```yaml
manifestVersion: 1.0

projects:
  - name: my-project
    path: projects/my-project

environmentGroups:
  - name: default
    environments:
      - name: development
        url:
          type: environment
          value: DT_DEV_URL
        auth:
          token:
            type: environment
            value: DT_DEV_TOKEN
      - name: production
        url:
          type: environment
          value: DT_PROD_URL
        auth:
          token:
            type: environment
            value: DT_PROD_TOKEN
```

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4. Configuration Files

> **About the schemas used in these examples.** Management zones, auto-tagging and alerting profiles appear throughout §4–§6 because they are the configurations most tenants already have, which makes them the clearest illustrations of Monaco's YAML shape. All three are marked **Blocked at upgrade** in AUTOM-02's Settings 2.0 catalog.
>
> That matters more for Monaco than for the raw API. Monaco resolves a schema before writing objects, so once a schema is removed, `monaco deploy` fails for every config that references it — and it fails at deploy time, in your pipeline, not at edit time. Treat these as **Monaco mechanics you can apply to any schema**, and check a schema's upgrade status before building a long-lived project around it.

> The `type.settings.schema` block is schema-agnostic: swap the schema string and the matching template JSON and everything else here is unchanged. Carry-forward schemas worth practising on include `builtin:davis.anomaly-detectors`, `builtin:anomaly-detection.services`, `builtin:synthetic.http`, and the `builtin:openpipeline.<scope>.*` family.

Basic Configuration Structure

```
configs:
  - id: production-zone
    config:
      name: Production
      template: management-zone.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment
```

Management Zone Example

config.yaml:

```
configs:
  - id: production-mz
    config:
      name: Production
      template: production-mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment

  - id: staging-mz
    config:
      name: Staging
      template: staging-mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment
```

production-mz.json:

```
{
  "name": "{{ .name }}",
  "rules": [
    {
      "type": "SERVICE",
      "enabled": true,
      "conditions": [
        {
          "key": {
            "type": "STATIC",
            "attribute": "SERVICE_TAGS"
          },
          "comparisonInfo": {
            "type": "TAG",
            "operator": "EQUALS",
            "value": {
              "context": "CONTEXTLESS",
              "key": "environment",
              "value": "production"
            },
            "negate": false
          }
        }
      ]
    }
  ]
}
```

Auto-Tagging Example

config.yaml:

```
configs:
  - id: application-tag
    config:
      name: Application
      template: application-tag.json
    type:
      settings:
        schema: builtin:tags.auto-tagging
        scope: environment
```

application-tag.json:

```
{
  "name": "{{ .name }}",
  "rules": [
    {
      "type": "SERVICE",
      "enabled": true,
      "valueFormat": "{Service:DetectedName}",
      "propagationTypes": ["SERVICE_TO_PROCESS_GROUP_LIKE"],
      "conditions": []
    }
  ]
}
```

Using Variables

Variables allow environment-specific values:

config.yaml:

```
configs:
  - id: alerting-profile
    config:
      name: "{{ .Env.ENVIRONMENT_NAME }} Alerting"
      template: alerting.json
      parameters:
        severity_level: "{{ .Env.ALERT_SEVERITY }}"
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

---

5. Common Operations

Deploy Configurations

Deploy to all environments:

```
monaco deploy manifest.yaml
```

Deploy to specific environment:

```
monaco deploy manifest.yaml --environment production
```

Dry run (validate without applying):

```
monaco deploy manifest.yaml --dry-run
```

Validate Configurations

Monaco does **not** ship a standalone `monaco validate` subcommand. Validation runs as part of `monaco deploy --dry-run`, which loads the manifest, parses every project, resolves cross-references, and contacts the target environment to confirm reachability and token scopes — all without making changes:

```
monaco deploy manifest.yaml --dry-run
```

Delete Configurations

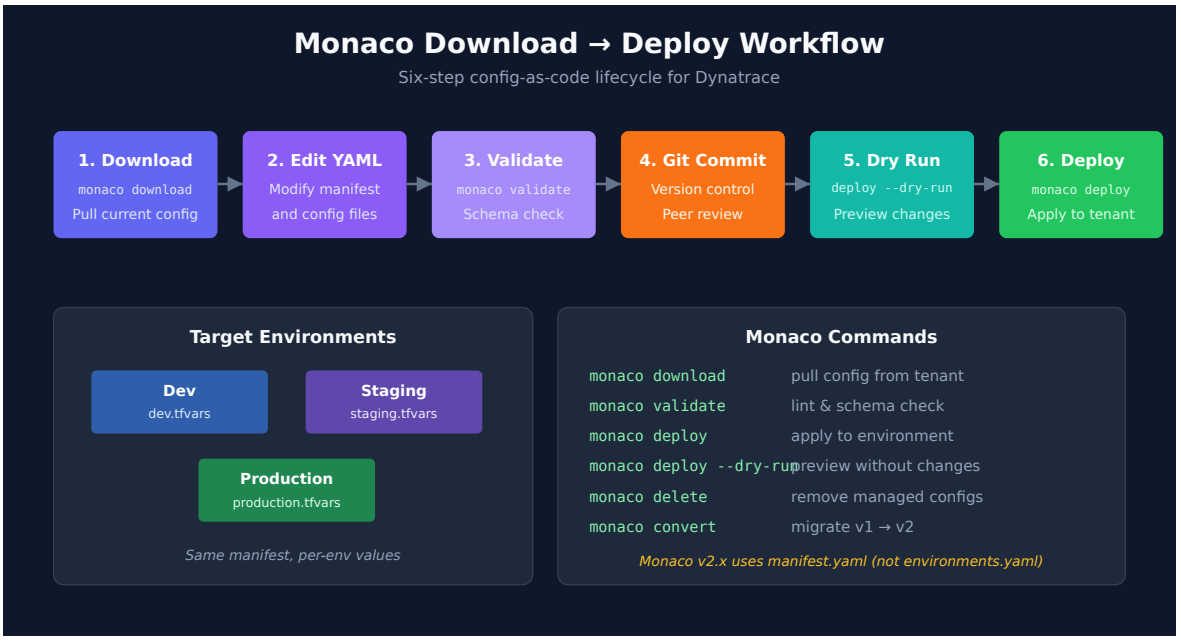
Remove configurations not in your project (orphans). The manifest is passed via `--manifest`, not as a positional argument:

```
monaco delete --manifest manifest.yaml --file delete.yaml
```

delete.yaml:

```
delete:
  - type: builtin:management-zones
    id: old-management-zone-id
```

Download vs Deploy Workflow



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6. Advanced Features

Configuration Dependencies

Reference other configurations:

```
configs:
  - id: my-alerting-profile
    config:
      name: Production Alerting
      template: alerting.json
      parameters:
        management_zone_id: "{{ .management-zones.production-mz.id }}"
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

Conditional Configurations

Skip configurations in certain environments:

```yaml
configs:
  - id: production-only-config
    config:
      name: Production Monitor
      template: monitor.json
      skip:
        environments:
          - development
          - staging
    type:
      settings:
        schema: builtin:synthetic.http
        scope: environment
```

Environment-Specific Overrides

```yaml
configs:
  - id: alerting
    config:
      name: Alerting Profile
      template: alerting.json
      parameters:
        delay_minutes: 5
      overrides:
        - environments:
            - production
          override:
            parameters:
              delay_minutes: 1
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

Grouping Configurations

Organize related configs in groups:

```
configs:
  - id: web-app-configs
    group: web-application
    config:
      name: Web App Management Zone
      template: mz.json
    type:
      settings:
        schema: builtin:management-zones
        scope: environment

  - id: web-app-alerting
    group: web-application
    config:
      name: Web App Alerting
      template: alerting.json
    type:
      settings:
        schema: builtin:alerting.profile
        scope: environment
```

Deploy only a specific group:

```
monaco deploy manifest.yaml --group web-application
```

Best Practices

Practice	Description
Use meaningful IDs	IDs should describe the config purpose
Modular structure	Separate configs by domain (alerting, monitoring, etc.)
Environment variables	Never hardcode tokens or URLs
Version control	Commit all changes with meaningful messages
Dry run first	Always validate before applying
Document parameters	Comment complex configurations

Practice	Description
Check upgrade status before you invest	Before building a long-lived project around a schema, confirm it is not marked **Blocked at upgrade** in AUTOM-02's catalog — a blocked schema takes the whole Monaco config type with it

Common Issues and Solutions

Issue	Cause	Solution
"Config already exists"	Duplicate ID	Use unique IDs or download first
"Schema not found"	Typo in schema ID — **or** the schema was removed when the tenant upgraded to the latest Dynatrace	Check the exact schema name in the API. If it was correct and worked previously, check the schema's upgrade status in AUTOM-02's catalog before assuming a typo
"Template not found"	Wrong path	Use relative path from config.yaml
"Validation failed"	Invalid JSON	Check JSON syntax in template

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7. Next Steps

Monaco in a CI/CD Pipeline (the most common next step)

Monaco itself is a CLI — to make it part of a GitOps flow, wire it into your CI/CD platform of choice. **AUTOM-07: CI/CD Integration** has the recipe per platform:

Platform	AUTOM-07 section
GitHub Actions	§3 GitHub Actions
GitLab CI/CD	§4 GitLab CI/CD
Bitbucket Pipelines	§5.1 Bitbucket Pipelines — Monaco Deploy

Platform	AUTOM-07 section
Atlassian Bamboo	§6 Atlassian Bamboo — adapt the Plan Specs YAML to call `monaco deploy`
Azure DevOps	§7 Azure DevOps Pipelines — adapt the Pipeline YAML to call `monaco deploy`

For the **full sequenced path** from zero to a working pipeline, see **AUTOM-01 §6 First-Time Setup Path — Path B (Monaco target)**.

When to Consider Adding Terraform

Scenario	Why Terraform wins
Cross-system orchestration (cloud + Dynatrace + Git in one apply)	Monaco is Dynatrace-only
State management + drift detection on critical resources	Monaco has no state file; drift detection happens via plan-reapply rather than state diffing
Part of larger IaC stack	Existing Terraform workflows extend naturally to Dynatrace
Lifecycle protections (`prevent_destroy` , `ignore_changes`)	Monaco has no per-resource lifecycle policy

For the framing of when a Terraform shop should *also* adopt Monaco (the reverse direction), see **AUTOM-01 §5: When a Terraform Shop Should Add Monaco**.

Continue the Series

Next Notebook	Focus
AUTOM-04: Terraform Provider	Infrastructure-as-code approach to Dynatrace configuration
AUTOM-07: CI/CD Integration	GitOps patterns for both Monaco and Terraform
AUTOM-08: Migration Automation	Tenant-to-tenant migration including `monaco download`

Additional Resources

- Monaco Documentation
- Monaco Examples
- Configuration Schema Reference

Summary

In this notebook, you learned:

- How to install and configure Monaco
- Project structure and manifest configuration
- Creating and deploying configurations via YAML
- Advanced features like dependencies, overrides, and groups

> **Key Takeaway:** Monaco is ideal for teams wanting GitOps-style configuration management. It provides the right balance of simplicity and power for most Dynatrace automation needs. For the from-zero-to-pipeline sequence, see **AUTOM-01 §6** Path B.

Continue to AUTOM-04: Terraform Provider to learn infrastructure-as-code patterns, or jump to AUTOM-07: CI/CD Integration to wire Monaco into a pipeline now.

Community Resources & Examples

The following GitHub repositories provide starter templates, real-world examples, and hands-on exercises for Monaco:

Official Repositories

Repository	Description
dynatrace-configuration-as-code	Official Monaco CLI (v2.28.x) -- Go binary, Apache-2.0
dynatrace-configuration-as-code-samples	Official samples repo with 9 Monaco starter templates in `basic-templates-monaco`
easytrade	Demo microservices app with a working `monaco/` directory (manifest.yaml, detection rules, workflows)
Dynatrace-Config-Manager	GUI tool for tenant-to-tenant config migration; complements Monaco for brownfield scenarios

Starter Templates (in `dynatrace-configuration-as-code-samples`)

Template Directory	What It Configures
`basic-templates-monaco`	Alerting, app detection, synthetic, maintenance window, management zones, ownership, notifications, SLOs
`learn-monaco-auto-tag`	Auto-tagging with Monaco
`account-monaco-admin-access`	Admin access setup for Monaco

Pipeline Observability Samples

Monaco configurations for ingesting CI/CD pipeline events via OpenPipeline:

Directory	CI/CD Platform
`github_pipeline_observability`	GitHub Actions
`gitlab_pipeline_observability`	GitLab CI
`azure_devops_observability`	Azure DevOps
`argocd_observability`	ArgoCD

Training & Exercises

Repository	Description
monaco-self-paced-exercises	6 structured exercises: install, auto-tag, download, variables, delete/restore, linking configs
monaco-demo	Working GitHub Actions workflow for Monaco deploy with "crawl-walk-run" adoption methodology

> **Note:** Monaco v1 (`dynatrace-oss/dynatrace-monitoring-as-code`) is no longer supported. Migrate v1 projects to v2 by re-running `monaco download` against the source tenant — there is no automated v1→v2 conversion subcommand.
